

Crimson Skies and Another Pale Blue Dot: Can Life Begin Again?

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INTRODUCTION

Enceladus is a primary target for extraterrestrial life, after the Cassini spacecraft flew by in 2005 and captured the remarkable jets in its southern pole. While Enceladus is currently a prime subject for prebiotic life, we investigate whether the evolution of our Sun from a main sequence star to a Red giant increases the odds of habitability for this icy moon.

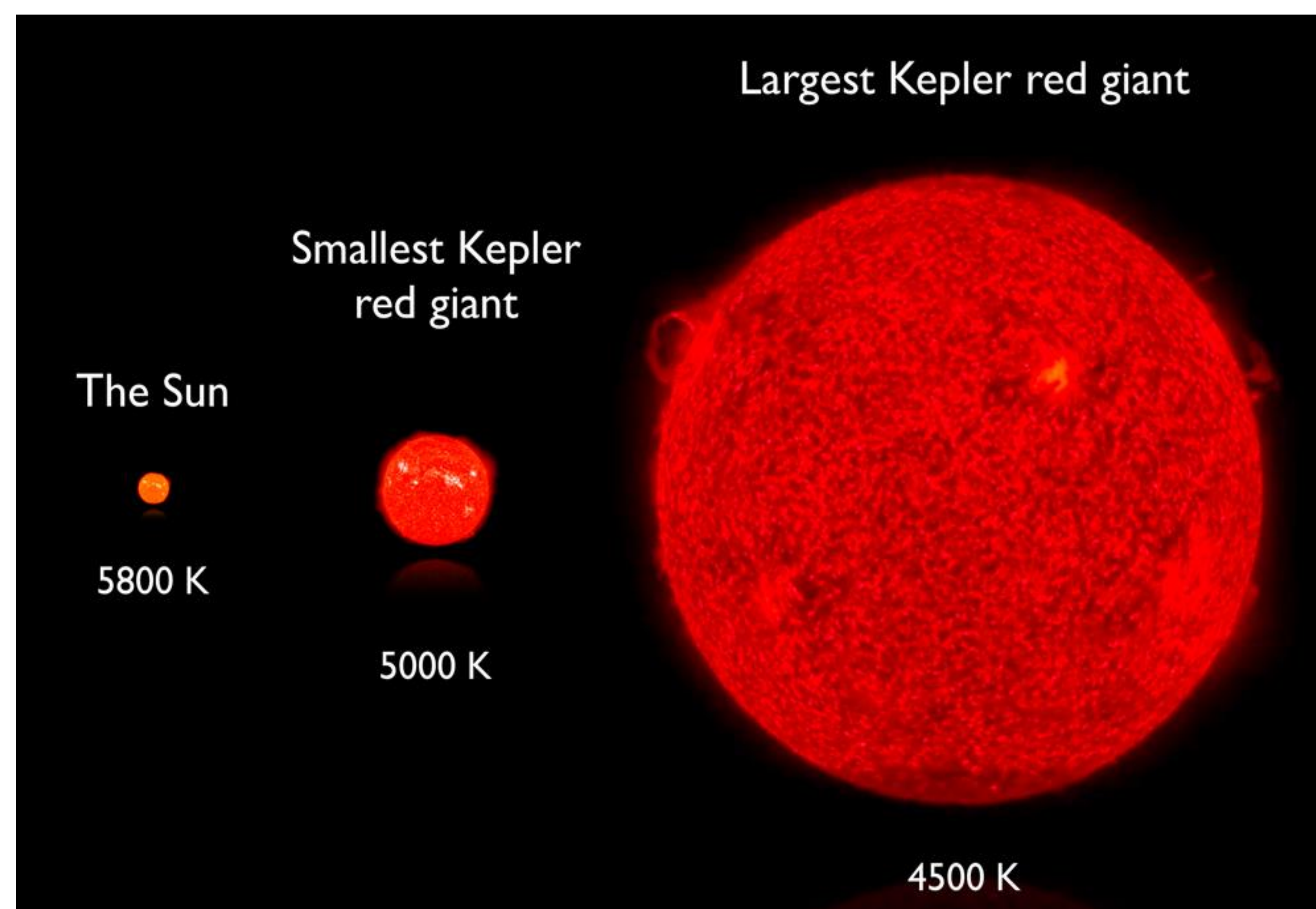


Figure 1: Evolutionary comparison between our current Sun and its predicted future.

METHODOLOGY

Using a Fortran based simulation modeling tool that functions in tandem with python, ExoPlaSim, was used to model the icy saturnian moon Enceladus. Running inside of terminal, we used our completely custom model parameters to run varied duration simulations. This was done to determine whether the increased flux from a transient main-sequence star would provide of enough energy to the distant moons of Saturn to sublime its thick ice shell creating liquid water and a habitability window through greenhouse effects of an exosphere. The primary alterations were made to flux of the parent star as well as atmospheric conditions, size, and gravity.

METHODOLOGY

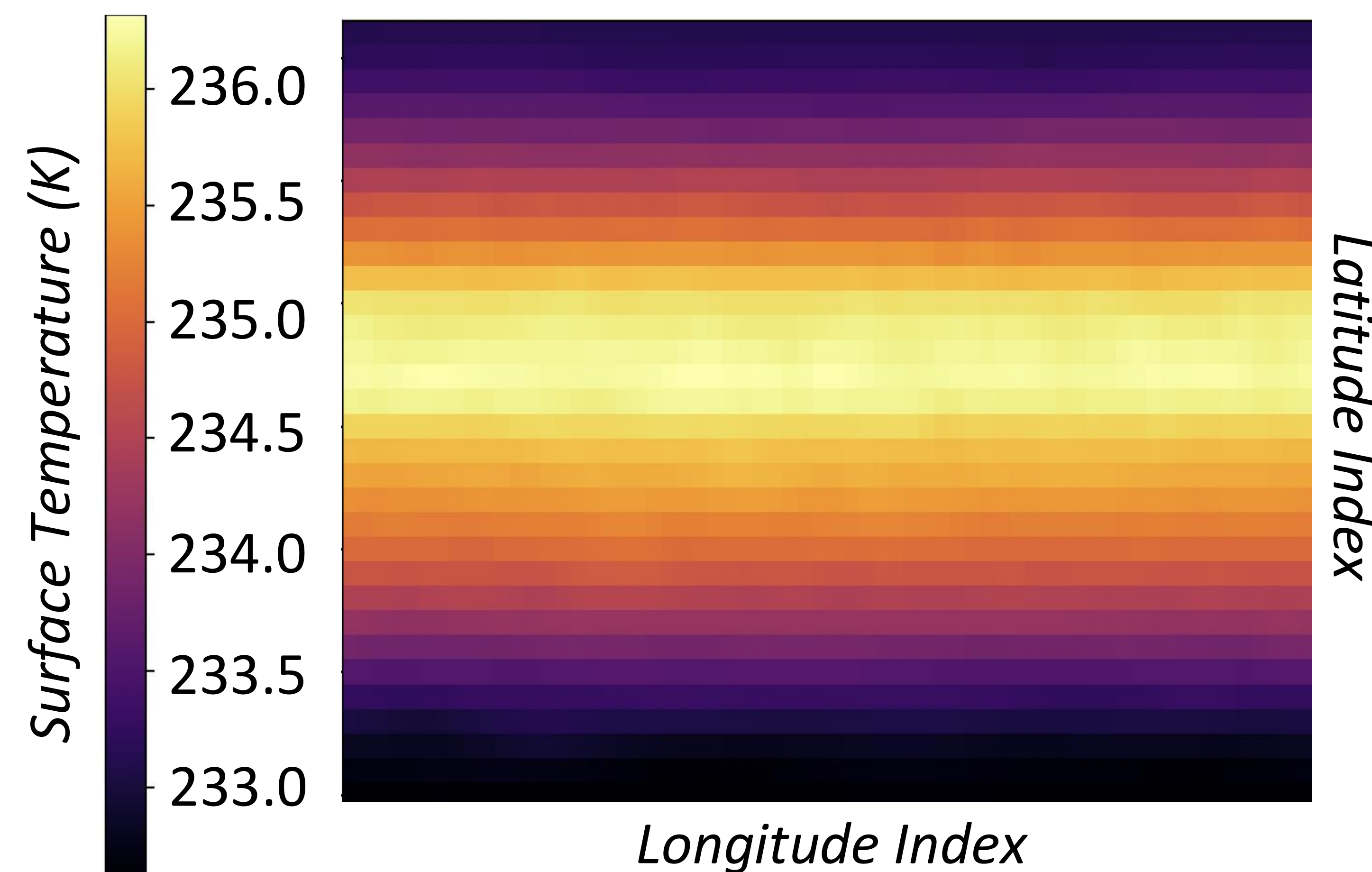


Figure 2: Enceladus Proxy Surface Temp. Global coordinate map under present-day solar conditions.

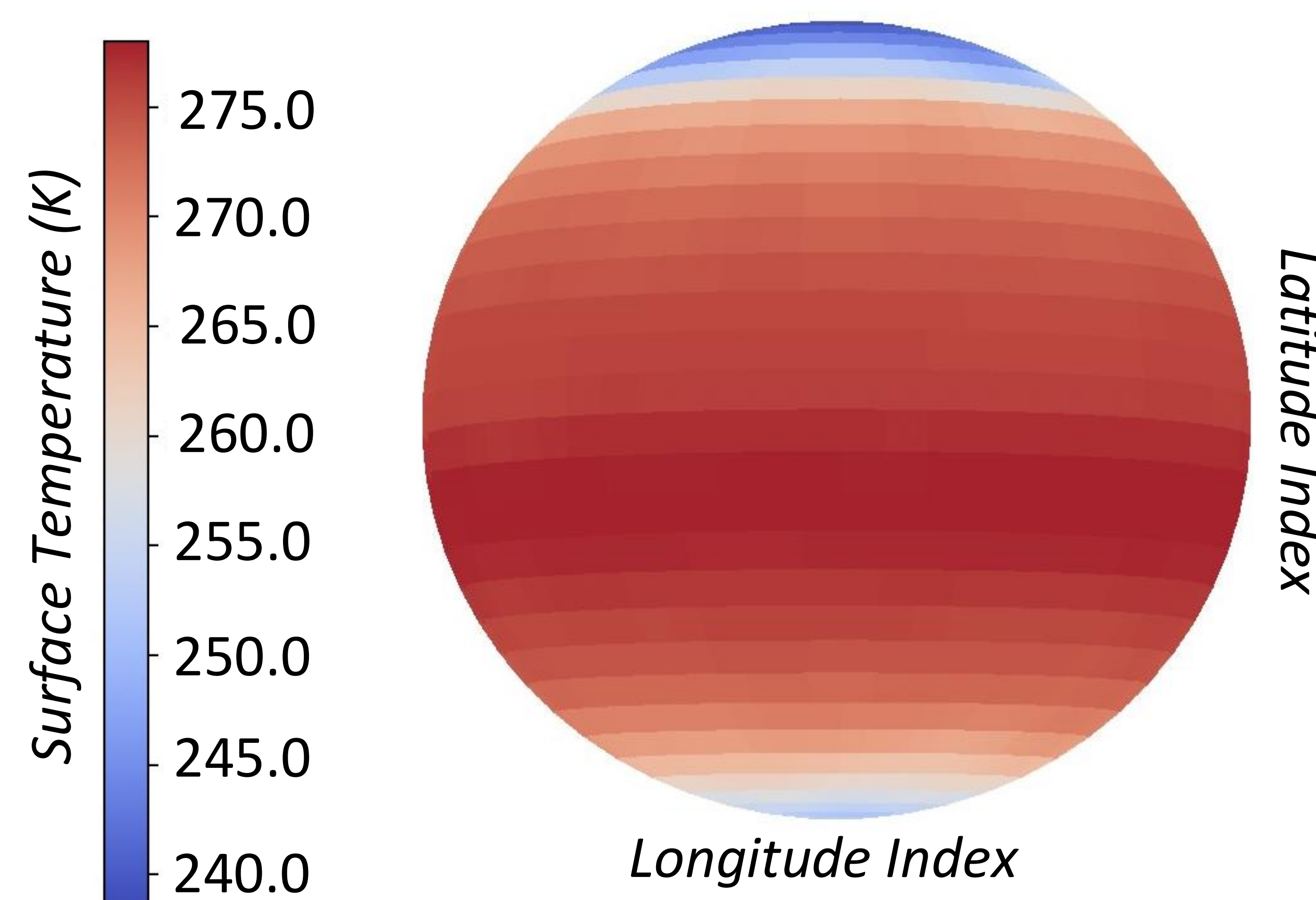


Figure 3: Surface Temp. under low atmospheric conditions and high flux from present solar values.

CONCLUSION

After thorough testing, it is found that the transition from Main Sequence to Red Giant will create a secondary habitable environment capable of increasing biological activity. These results provide crucial predictive data for future missions to icy moons, suggesting that post-main sequence stars may create transient habitable zones.

RESULTS

While our simulations are still in progress, we have theorized the results of our simulations. During the Main Sequence phase, habitability will remain confined to the subsurface ocean. However, as the Sun expands and the increase in radiative heating reaches Enceladus, we predict a second habitability zone will be created. As the Sun transitions into the final White Dwarf phase, the tiny icy moon will lose all potential for life in a “deep freeze”.

Through preliminary simulations, we were able to retrieve the following formula for the relationship between surface temperature and increasing stellar flux: $y=138x^{(0.096)}$.

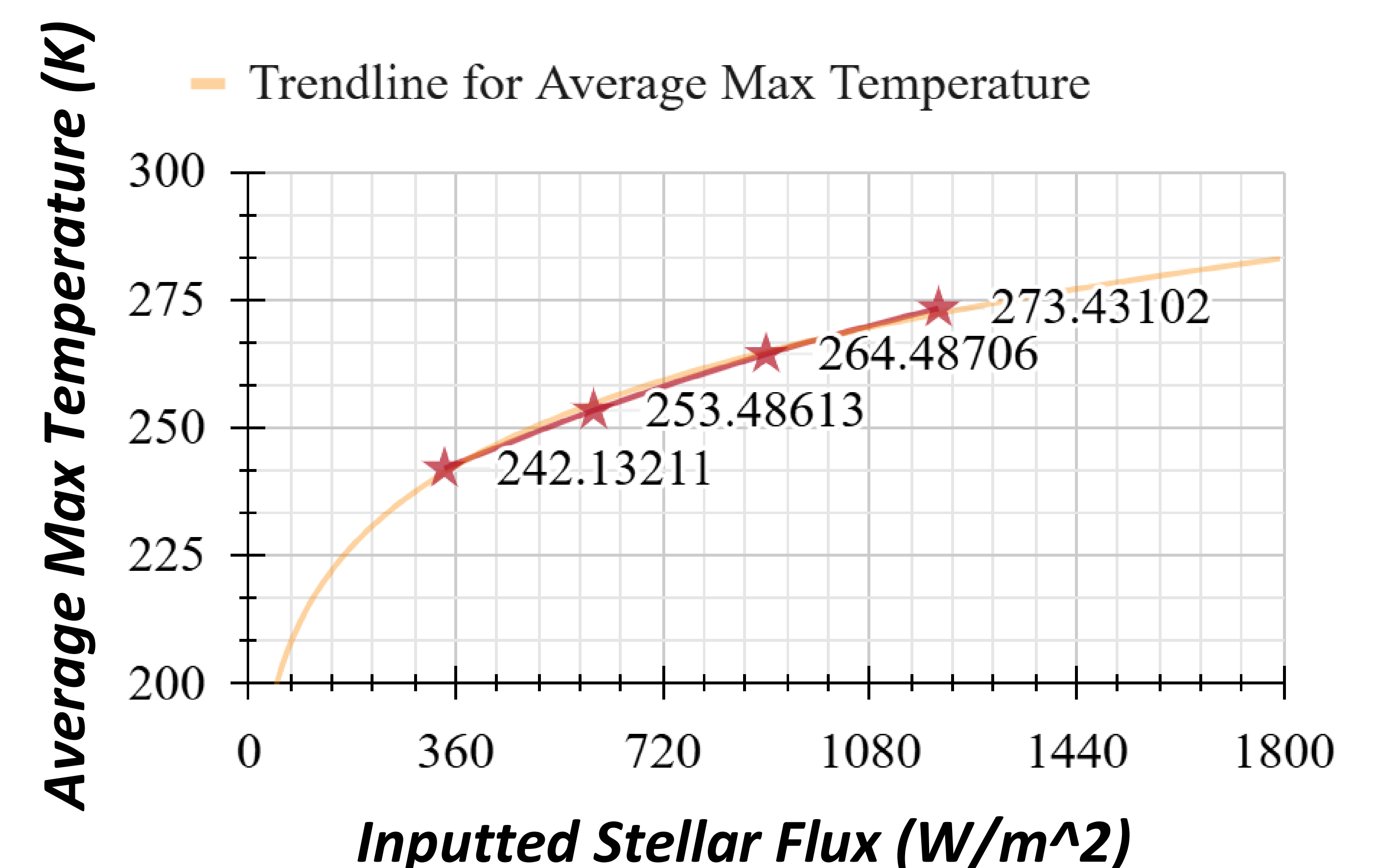


Figure 4: Simulated average max temperatures at various stellar fluxes.

REFERENCES

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